

Envelope Equations

USPAS - Beam Physics with Intense Space Charge

Austin Hoover Steven Lund Daniel Winklehner John Barnard

July 2026

Paraxial ray equation

Envelope equations for axially symmetric beams

Envelope equations for elliptically symmetric beams

Envelope equations: reduced-order characterization of beam dynamics in terms of second-order moments

[Image]

Road map

- Start from single particle equation of motion: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
- Make use of:
 - Paraxial (near-axis) approximation
 - Conservation of angular momentum
 - Axisymmetry of external forces
- Arrive at single-particle paraxial ray equation.
- Take statistical averages to obtain moment/envelope equations.

Paraxial ray equation

Equation of motion in cylindrical coordinates

Equation of motion for particle of mass m and charge q in electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$:

$$\frac{d\mathbf{p}}{dt} = \frac{d(\gamma m \dot{\mathbf{x}})}{dt} = q(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}), \quad (1)$$

Switch to cylindrical coordinates (r, θ, z) :

$$\begin{aligned} \mathbf{p} &= p_x \hat{x} + p_y \hat{y} + p_z \hat{z} \\ &= p_r \hat{r} + p_\theta^* \hat{\theta} + p_z \hat{z}. \end{aligned} \quad (2)$$

The components p_r and p_θ^* are given by

$$\begin{aligned} p_r &= \gamma m \dot{r}, \\ p_\theta^* &= \gamma m r \dot{\theta}. \end{aligned} \quad (3)$$

The symbol p_θ^* is used to reserve p_θ for the *canonical* angular momentum, defined later.

In cylindrical coordinates, the unit vectors depend on the particle position and thus change with time:

$$\begin{aligned}\dot{\hat{r}} &\equiv d\hat{r}/dt = \dot{\theta}\hat{\theta}, \\ \dot{\hat{\theta}} &\equiv d\hat{\theta}/dt = -\dot{\theta}\hat{r}.\end{aligned}\tag{4}$$

Thus, the time derivative of the momentum vector in Eq. (2) is:

$$\frac{d\mathbf{p}}{dt} = \frac{d}{dt} (p_r\hat{r} + p_\theta^*\hat{\theta} + p_z\hat{z})\tag{5}$$

$$= \dot{p}_r\hat{r} + p_r\dot{\hat{r}} + \dot{p}_\theta\hat{\theta} + p_\theta^*\dot{\hat{\theta}} + \dot{p}_z\hat{z}\tag{6}$$

$$= (\dot{p}_r - p_\theta^*\dot{\theta})\hat{r} + (\dot{p}_\theta + p_r\dot{\theta})\hat{\theta} + \dot{p}_z\hat{z}\tag{7}$$

$$= \left[\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 \right] \hat{r} + \left[\frac{d(\gamma m r \dot{\theta})}{dt} + \gamma m \dot{r} \dot{\theta} \right] \hat{\theta} + \left[\frac{d(\gamma m \dot{z})}{dt} \right] \hat{z}.\tag{8}$$

$$= \left[\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 \right] \hat{r} + \left[\frac{1}{r} \frac{d}{dt} (\gamma m r^2 \dot{\theta}) \right] \hat{\theta} + \left[\frac{d(\gamma m \dot{z})}{dt} \right] \hat{z}.\tag{9}$$

This takes care of the left side of the Lorentz force equation (Eq. (1)).

The right side of Eq. (1) requires the cylindrical field components:

$$\begin{aligned}\mathbf{E} &= E_r \hat{r} + E_\theta \hat{\theta} + E_z \hat{z}, \\ \mathbf{B} &= B_r \hat{r} + B_\theta \hat{\theta} + B_z \hat{z}.\end{aligned}\tag{10}$$

Evaluating the cross product $\dot{\mathbf{x}} \times \mathbf{B}$,

$$\dot{\mathbf{x}} \times \mathbf{B} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ \dot{r} & r\dot{\theta} & \dot{z} \\ B_r & B_\theta & B_z \end{vmatrix},\tag{11}$$

leads to

$$\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 = q (E_r + r \dot{\theta} B_z - \dot{z} B_\theta),\tag{12}$$

$$\frac{1}{r} \frac{d(\gamma m r^2 \dot{\theta})}{dt} = q (E_\theta + \dot{z} B_r - \dot{r} B_z),\tag{13}$$

$$\frac{d(\gamma m \dot{z})}{dt} = q (E_z + \dot{r} B_\theta - r \dot{\theta} B_r)\tag{14}$$

From fields to potentials

Write the fields in terms of the scalar potential $\phi(\mathbf{x})$ and vector potential $\mathbf{A}(\mathbf{x})$:

$$\begin{aligned}\mathbf{E} &= -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \\ \mathbf{B} &= \nabla \times \mathbf{A}.\end{aligned}\tag{15}$$

The gradient $\nabla\phi$ in cylindrical coordinates is:

$$\nabla\phi = \left[\frac{\partial\phi}{\partial r}\right]\hat{r} + \left[\frac{1}{r}\frac{\partial\phi}{\partial\theta}\right]\hat{\theta} + \left[\frac{\partial\phi}{\partial z}\right]\hat{z}.\tag{16}$$

And the curl $\nabla \times \mathbf{A}$ is:

$$\nabla \times \mathbf{A} = \left[\frac{1}{r}\frac{\partial A_z}{\partial\theta} - \frac{\partial A_\theta}{\partial z}\right]\hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\hat{\theta} + \frac{1}{r}\left[\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right]\hat{z}.\tag{17}$$

Axial symmetry means all partial derivatives with respect to θ are zero in Eq. (16) and Eq. (17). This gives the following expressions for the fields in terms of the potentials in cylindrical coordinates:

$$\begin{aligned}\mathbf{E} &= \left[-\frac{\partial\phi}{\partial r} - \frac{\partial A_r}{\partial t} \right] \hat{r} + \left[-\frac{\partial A_\theta}{\partial t} \right] \hat{\theta} + \left[-\frac{\partial\phi}{\partial z} - \frac{\partial A_z}{\partial t} \right] \hat{z}, \\ \mathbf{B} &= \left[-\frac{\partial A_\theta}{\partial r} \right] \hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right] \hat{\theta} + \left[\frac{1}{r} \frac{\partial(rA_\theta)}{\partial r} \right] \hat{z}.\end{aligned}\tag{18}$$

Conservation of angular momentum

Use the potentials to rewrite the angular equation of motion (Eq. (13)):

$$\begin{aligned}\frac{d}{dt}(\gamma m r^2 \dot{\theta}) &= q r (E_{\theta} + \dot{z} B_r - \dot{r} B_z) \\&= q r \left(\left[-\frac{\partial A_{\theta}}{\partial t} \right] + \dot{z} \left[-\frac{\partial A_{\theta}}{\partial z} \right] - \dot{r} \left[\frac{1}{r} \frac{\partial(r A_{\theta})}{\partial r} \right] \right) \\&= -q \left(\frac{\partial(r A_{\theta})}{\partial t} + \dot{z} \frac{\partial(r A_{\theta})}{\partial z} + \dot{r} \frac{\partial(r A_{\theta})}{\partial r} \right) \\&= -q \left[\frac{\partial}{\partial t} + \dot{z} \frac{\partial}{\partial z} + \dot{r} \frac{\partial}{\partial r} \right] (r A_{\theta}) \\&= -q \left[\frac{\partial}{\partial t} + \dot{\mathbf{x}} \cdot \nabla \right] (r A_{\theta}) \\&= -\frac{d(q r A_{\theta})}{dt}.\end{aligned}\tag{19}$$

Eq. (19) can then be written as:

$$\frac{d}{dt} (\gamma m r^2 \dot{\theta} + q r A_{\theta}) = 0. \quad (20)$$

This motivates the definition of the *canonical angular momentum*:

$$p_{\theta} \equiv \gamma m r^2 \dot{\theta} + q r A_{\theta}, \quad (21)$$

such that $dp_{\theta}/dt = 0$. We may also write p_{θ} in terms of the magnetic flux $\psi(r)$ though a circle of radius r [Figure]:

$$\psi(r) = \int \mathbf{B} \cdot d\Omega = \int \nabla \times \mathbf{A} \cdot d\Omega = \oint \mathbf{A} \cdot d\mathbf{l} = 2\pi r A_{\theta}. \quad (22)$$

Therefore, Eq. (21) can be written

$$p_{\theta} \equiv \gamma m r^2 \dot{\theta} + \frac{q\psi(r)}{2\pi}. \quad (23)$$

Series expansion of axially symmetric fields

Consider external fields $\mathbf{E}$ and $\mathbf{B}$ with azimuthal symmetry ($\partial/\partial\theta = 0$). In the absence of charge and current densities,

$$\begin{aligned}\nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= 0, \\ \nabla \times \mathbf{B} &= 0.\end{aligned}\tag{24}$$

The vanishing curl means we can express the fields as

$$\begin{aligned}\mathbf{E} &= -\nabla\phi, \\ \mathbf{B} &= -\nabla\phi_m,\end{aligned}\tag{25}$$

for scalar potentials ϕ and ϕ_m . Additionally, the vanishing divergence leads to the Laplace equation for the scalar potential:

$$\nabla^2\phi = 0.\tag{26}$$

In cylindrical coordinates, the Laplace equation reads:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial^2 \phi}{\partial z^2} = 0. \quad (27)$$

The azimuthal symmetry will lead to a simplified form of the scalar potential in terms of its on-axis derivatives along the z axis. First, expand the potential $\phi(r, z)$ as a power series,

$$\phi(r, z) = \sum_{n=0}^{\infty} h_{2n}(z) r^{2n}, \quad (28)$$

where the $h_{2n}(z)$ are functions of z . (The rotational symmetry eliminates all odd powers of r .) The derivatives in the Laplace equation are:

$$\begin{aligned} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) &= \sum_{n=1}^{\infty} (2n)^2 h_{2n}(z) r^{2n-2}, \\ \frac{\partial^2 \phi}{\partial z^2} &= \sum_{n=0}^{\infty} h'_{2n}(z) r^{2n}, \end{aligned} \quad (29)$$

with $h'(z) = dh(z)/dz$.

Thus, $\nabla^2\phi = 0$ becomes:

$$\begin{aligned}
\sum_{n=1}^{\infty} (2n)^2 h_{2n}(z) r^{2n-2} + \sum_{n=0}^{\infty} h_{2n}''(z) r^{2n} &= 0 \\
\sum_{n=0}^{\infty} (2n+2)^2 h_{2n+2}(z) r^{2n} + \sum_{n=0}^{\infty} h_{2n}''(z) r^{2n} &= 0 \\
\sum_{n=0}^{\infty} [(2n+2)^2 h_{2n+2}(z) + h_{2n}''(z)] &= 0
\end{aligned} \tag{30}$$

The result is a recurrence relation,

$$h_{2n+2}(z) = -\frac{h_{2n}''(z)}{(2n+2)^2}, \tag{31}$$

with which we can write all terms as derivatives of $h_0(z)$ with respect to z :

$$\begin{aligned}
h_2(z) &= -\frac{1}{4} \frac{\partial^2 h_0(z)}{\partial z^2}, \\
h_4(z) &= +\frac{1}{64} \frac{\partial^4 h_0(z)}{\partial z^4}, \\
&\vdots
\end{aligned} \tag{32}$$

We may thus write the potential at all points in space using only the z -derivatives of the on-axis potential $\phi(0, z)$:

$$\phi(r, z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n} \phi(0, z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n}. \quad (33)$$

We now express the fields in terms of the expanded scalar potentials. Write the magnetic field as $\mathbf{B} = B_r \hat{r} + B_z \hat{z}$. Using the series expansion of $\phi_m(r, z)$, we find for the z component of the field:

$$\begin{aligned}
B_z(r, z) &= -\frac{\partial \phi_m}{\partial z} \\
&= -\frac{\partial}{\partial z} \left[\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n} \phi_m(0, z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^2 \right] \\
&= -\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n+1} \phi_m(0, z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^2.
\end{aligned} \tag{34}$$

Define $B(z)$ as the on-axis value of $B_z(z)$:

$$B(z) \equiv B_z(0, z) = -\frac{\partial \phi_m(0, z)}{\partial z}. \tag{35}$$

Inserting Eq. (35) into Eq. (34) gives:

$$\begin{aligned}
B_z(r, z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n}}{\partial z^{2n}} \left[-\frac{\partial \phi_m(0, z)}{\partial z} \right] \left(\frac{r}{2} \right)^2 \\
B_z(r, z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n} B(z)}{\partial z^{2n}} \left(\frac{r}{2} \right)^2.
\end{aligned} \tag{36}$$

For the radial component, we find:

$$B_r(r, z) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n-1)!} \frac{\partial^{2n-1} B(z)}{\partial z^{2n-1}} \left(\frac{r}{2} \right)^{2n-1}. \tag{37}$$

A similar procedure gives the radial and longitudinal components of the electric field $\mathbf{E} = E_r \hat{r} + E_z \hat{z}$. Define $V(z)$ as the on-axis value of the electric potential ϕ :

$$V(z) \equiv \phi(0, z). \quad (38)$$

This gives:

$$\begin{aligned} E_z(r, z) &= -\frac{\partial \phi}{\partial z} = -\sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n+1} V(z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^{2n}. \\ E_r(r, z) &= -\frac{\partial \phi}{\partial r} = -\sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n-1)!} \frac{\partial^{2n} V(z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n-1}. \end{aligned} \quad (39)$$

Derivation of paraxial ray equation

We will now take the *paraxial* limit of small transverse velocity relative to longitudinal velocity: $\dot{r} \ll \dot{z}$, $r\dot{\theta} \ll \dot{z}$. If $v = (v_r^2 + v_\theta^2 + v_z^2)^{1/2}$ is the total velocity, where $v_r = \dot{r}$, $v_\theta = r\dot{\theta}$, and $v_z = \dot{z}$, the paraxial limit gives $\dot{z} = (v^2 - \dot{r}^2 - r^2\dot{\theta}^2)^{1/2} \approx v$. We will therefore only keep terms linear in r in the equations of motion.

We will also add the self-generated fields to the equation of motion, writing

$$\begin{aligned}\mathbf{E} &= \mathbf{E}^{(ext)} + \mathbf{E}^{(self)}, \\ \mathbf{B} &= \mathbf{B}^{(ext)} + \mathbf{B}^{(self)}.\end{aligned}\tag{40}$$

[Is there a way to derive the ray equation first, then add the self-fields later?]

To first order in r , the external fields are:

$$\begin{aligned}
E_r^{(ext)} &\approx \frac{r}{2} \frac{\partial^2 V(z)}{\partial z^2}, \\
E_z^{(ext)} &\approx -\frac{\partial V(z)}{\partial z}, \\
B_r^{(ext)} &\approx -\frac{r}{2} \frac{\partial B(z)}{\partial z}, \\
B_z^{(ext)} &\approx B(z).
\end{aligned} \tag{41}$$

Recall that $V(z) = \phi(0, z)$ and $B(z) = -\partial\phi_m(0, z)/\partial z$.

Plug these truncated fields into Eq. (??) (the radial equation of motion):

$$\begin{aligned}
\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 &= q \left(E_r + r \dot{\theta} B_z - \dot{z} B_\theta \right), \\
&\approx q \underbrace{\left(\frac{r}{2} \frac{\partial^2 V(z)}{\partial z^2} + r \dot{\theta} B(z) \right)}_{\text{external fields}} + q \underbrace{\left(E_r^{(self)} - \dot{z} B_\theta^{(self)} \right)}_{\text{self fields}}.
\end{aligned} \tag{42}$$

The $B_z^{(self)}$ term is ignored in the paraxial limit.

Two more steps will lead to our definition of the *paraxial ray equation*. First, we will switch to the axial coordinate z as the independent variable rather than the time t . It is more common in accelerator physics to use the label s , so we define $s \equiv z$ and $ds = v_z dt = \beta c dt$, where c is the speed of light. Also define $' \equiv d/ds$. This gives (dropping the $\approx$ symbol from now on):

$$\beta c \frac{d(\gamma \beta m c r')}{ds} - \gamma m r (\beta c \theta')^2 = q \left(\frac{1}{2} r V''(s) + r \beta c \theta' B(s) \right) + q \left(E_r^{(self)} - \beta c B_\theta^{(self)} \right). \quad (43)$$

Expand the derivative in the first term and divide by $\gamma m \beta^2 c^2$:

$$r'' + \frac{(\gamma \beta)'}{(\gamma \beta)} r' - r \theta'^2 = \frac{q}{\gamma m \beta^2 c^2} \left(\frac{r}{2} V''(s) + \beta c r \theta' B(s) + E_r^{(self)} - \beta c B_\theta^{(self)} \right). \quad (44)$$

Now use the definition of the canonical angular momentum p_θ (Eq. (21)) to eliminate θ' , noting that the flux $\psi(r) = \pi r^2 B$:

$$\begin{aligned}\theta' &= \frac{\dot{\theta}}{\beta c} \\ \theta' &= \frac{1}{\gamma \beta m c r^2} \left(p_\theta - \frac{q \psi(r)}{2\pi} \right), \\ \theta' &= \frac{p_\theta}{\gamma \beta m c r^2} - \frac{\omega_c}{2\gamma \beta c},\end{aligned}\tag{45}$$

where we have defined the *cyclotron frequency*:

$$\omega_c \equiv \frac{qB}{m}.\tag{46}$$

Substitution of Eq. (45) into Eq. (??) gives:

$$\begin{aligned}r'' + \frac{(\gamma\beta)'}{\gamma\beta} r + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 r - \left(\frac{p_\theta}{\gamma\beta m c} \right)^2 \frac{1}{r^3} = \\ \frac{q}{\gamma m \beta^2 c^2} \left(\frac{V''}{2} r + E_r^{(self)} - \beta c B_\theta^{(self)} \right)\end{aligned}\tag{47}$$

Let us derive simplified expressions for $E_r^{(self)}$ and $B_\theta^{(self)}$ in terms of the charge and current density. Assuming cylindrical symmetry of the beam, the Poisson equation for the self-generated electric field is:

$$\nabla^2 \phi^{(self)} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi^{(self)}}{\partial r} \right) + \frac{\partial^2 \phi^{(self)}}{\partial s^2} = -\frac{\rho}{\epsilon_0}. \quad (48)$$

{#eq-poisson-self-field}

Solving for $\partial \phi^{(self)} / \partial r$:

$$E_r^{(self)} = -\frac{\partial \phi^{(self)}}{\partial r} = \frac{\lambda(r)}{2\pi\epsilon_0 r} + \frac{r}{2} \frac{\partial^2 \phi^{(self)}}{\partial s^2}, \quad (49)$$

where λ is the line-charge density:

$$\lambda(r) \equiv 2\pi \int_0^r \rho(\tilde{r}) \tilde{r} d\tilde{r}, \quad (50)$$

For the $B_{\theta}^{(self)}$ component of the self-generated magnetic field, we use Ampere's law:

$$2\pi r B_{\theta}^{(self)} = \mu_0 \int_0^r 2\pi \tilde{r} J_z(\tilde{r}) d\tilde{r} = \mu_0 \beta c \lambda(r) \quad (51)$$

Using the definition $\mu_0 \epsilon_0 = c^{-2}$,

$$B_{\theta}^{(self)} = \frac{\beta}{c} \frac{\lambda(r)}{2\pi \epsilon_0 r}. \quad (52)$$

With these approximations of the self-field components, Eq. (??) becomes:

$$r'' + \frac{(\gamma\beta)'}{\gamma\beta} r + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 r - \left(\frac{p_{\theta}}{\gamma\beta m c} \right)^2 \frac{1}{r^3} = \frac{q}{\gamma m \beta^2 c^2} \left[\frac{r}{2} \left(\frac{\partial^2 V}{\partial s^2} + \frac{\partial^2 \phi^{(self)}}{\partial s^2} \right) + \frac{1}{\gamma^2} \frac{\lambda(r)}{2\pi \epsilon_0 r} \right]. \quad (53)$$

A further simplification can be made by for the second derivatives of the potentials. The rate of change in the particle energy is given by

$$\frac{d(\gamma mc^2)}{dt} = q\mathbf{E} \cdot \mathbf{v}. \quad (54)$$

In the paraxial limit, the energy gain is almost entirely due to the longitudinal field:

$$\frac{d(\gamma mc^2)}{dt} \approx qE_z v_z = q\beta c E_z. \quad (55)$$

Switching to s as the independent variable, we find

$$\begin{aligned} \gamma' &\approx \frac{qE_z}{mc^2} \\ \gamma'' &\approx \frac{q}{mc^2} \frac{dE_z}{ds} \\ \gamma'' &\approx \frac{q}{mc^2} \frac{d}{ds} \left(E_z^{(ext)} + E_z^{(self)} \right) \\ \gamma'' &\approx -\frac{q}{mc^2} \left(\frac{\partial^2 V}{\partial s^2} + \frac{\partial^2 \phi^{(self)}}{\partial s^2} \right). \end{aligned} \quad (56)$$

Inserting this approximation of λ'' in Eq. (??) gives our final form of the *paraxial ray equation*:

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)}r' + \left(\frac{\gamma''}{2q\gamma\beta^2}\right)r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma\beta mc}\right)^2 \frac{1}{r^3} = \frac{q}{mc^2\beta^2\gamma^3} \frac{\lambda(r)}{2\pi\epsilon_0 r}} \quad (57)$$

The equations are valid for axisymmetric external focusing forces in the paraxial limit, with the inclusion of linearized self-fields of an axially symmetric beam. The terms in Eq. (??), in order: (i) [...]; (ii) accelerative damping; E_r from converging field lines; (iii) solenoidal focusing ($v_\theta B_z$); (iv) centrifugal defocusing; (v) linearized self-field ($E_r - v_z B_\theta$).

Envelope equations for axially symmetric beams

Envelope equations for axially symmetric beams

The strategy in this section will be to compute the second moments of the transverse distribution function $f(x, x', y, y', s)$. For axially symmetric beams, the Vlasov equation will generate a set of differential equations for the evolution of the second moments such as $\langle rr \rangle$, $\langle rr' \rangle$, $\langle rr'' \rangle$, etc. Combining with the paraxial ray equation will generate a set of ordinary differential equations (ODEs) for the evolution of the beam *envelope* radius $r_b = \sqrt{2\langle rr \rangle}$.

Start with the Vlasov equation:

$$\frac{\partial f}{\partial s} + \mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} + \mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} = 0, \quad (58)$$

where $\mathbf{x} = (x, y)$. Multiply the Vlasov equation by $g(\mathbf{x}, \mathbf{x}')$, an arbitrary function of the phase space coordinates, and integrate over all phase space space:

$$\iint \left[g \frac{\partial f}{\partial s} + g \mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} + g \mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} \right] d\mathbf{x} d\mathbf{x}' = 0. \quad (59)$$

Introduce bracket notation to indicate the expected value under the distribution; e.g.,

Envelope equations for elliptically symmetric beams
